Optimal Coherent Processing Interval Selection for Aerial Maneuvering Target Imaging Using Tracking Information

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***Abstract*— When a target is in complex three-dimensional (3-D) motions, inverse synthetic aperture radar (ISAR) images cannot be obtained with any motion compensation algorithms. In this case, selecting optimal coherent processing interval (CPI) (or frame) of the fixed rotational axis is the only way to obtain focused ISAR images. In this article, we focus on selecting optimal CPI for ISAR imaging of maneuvering targets, and a novel method using tracking information is proposed. Without complex computational procedures, the proposed method provides robust performance. The method includes three steps. First, target's orientation (roll, pitch, and yaw) is estimated by using an extended Kalman filter. Second, the least-square method is adopted to find the best fitted lines of roll, pitch, and yaw angles in different time intervals, and the linearity degrees of the three rotation angles are measured over the imaging intervals. Third, the thresholds of the linearity degrees are set, according to which, optimal CPIs can be selected. The performance of proposed method is demonstrated using both simulated data and measured data. In both cases, optimal CPIs can be selected and well-focused ISAR images can be achieved.**

***Index Terms*— Inverse synthetic aperture radar (ISAR), maneuvering target, coherent processing interval, tracking information.**

I. Introduction

INVERSE synthetic aperture radar (ISAR) imaging has wide applications in military and civil areas due to the ability of obtaining 2-D high resolution images of non-cooperative moving targets at a long distance under all weather and all day [1], [2]. In an ISAR imaging system, motion of a maneuvering target can be always decomposed into two parts: translational motion (TM) and rotational motion (RM). TM is undesired because it induces inevitable blurring of ISAR images. Therefore, TM compensation should be performed to obtain focused ISAR images [3]–[12]. In contrast, RM, which causes a change in direction of the aspect angle between a target and the radar's line of sight (LOS), is desired to separate scatterers in the cross-range direction. Although RM is an essential motion component for ISAR imaging, ISAR images are blurred when the rate of change of RM is not constant during the coherent processing interval (CPI). In this case, additional RM compensation algorithms [13]–[19] should be employed to form focused ISAR images. These methods work well when the rotational axis of a target stays constant during CPI, such that the RM of the target is confined to a 2D image projection plane (IPP), this type of rotation is referred to as 2D RM. However, If target is maneuvering during the CPI, such as maneuvering target, the complexity of the 3-D motion (roll, pitch, and yaw) induces strong time-varying characteristics of IPP [20]–[22]. Consequently, the Doppler frequency associated with each scatter would become time-varying, and thus, the ISAR image of the target may be degraded by using the aforementioned motion compensation algorithms. For ISAR imaging of maneuvering targets, another option is selecting an optimum CPI to balance the focus quality and resolution for the ISAR image. In this paper, we mainly focus on the optimum CPI selection to improve imaging success rate and image quality.

A number of works have been developed on optimum CPI selection for ISAR imaging, which usually improve the quality and feasibility of ISAR imaging for a target with complex rotational motion. The existing algorithms can be generally sorted into two categories. The first category is established based on signal or image analysis [23]–[25], which first forms ISAR images and utilizes the images features to determine the maneuvering severity of the target, and the optimum CPIs can be obtained without any help of the priori knowledge of target motion. Their shortcoming lies in the low efficiency for image analysis. The second category is based on Doppler frequency analysis [26]–[31]. They rely on extracting the Doppler frequencies of scattering points to estimate the target motion characteristics. Literature [28] introduced an optimum CPIs selection method based on Doppler Centroid Frequency (DCF), which involves Doppler tracking of prominent scattering centers. In this category, the time-frequency analysis for instantaneous frequency estimation is usually applied. The methods in [29] and [30] select the CPIs, within which, the instantaneous Doppler frequency is approximately constant. They usually need to obtain the optimal kernel which

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would increase the computational complexity. In [32] and [33] adaptive joint time-frequency (AJTF) projection algorithm is used to determine the phase signals of prominent scatterers, and the optimum frames are determined by measuring the nonlinearity of scatterers' phase. However, the noise always influences the accuracy of the phase estimation. Moreover, these existing selection methods mainly concentrate on ISAR imaging of marine targets by using data itself, which suffers from the low quality of the received data (the received data is always interfered with noise or sea clutter). Besides, rare algorithms have been established to select optimal CPI for aerial targets.

The core idea of the proposed method in this paper is to take the advantage of the tracking information to obtain the time-varying orientation (pitch, roll and yaw) of an aerial target. Based on the orientation information, the optimum CPIs can be optimally determined. The algorithm consists of three steps. Firstly, target's velocity is calculated by using tracking information, then, the three rotational components, namely pitch, roll and yaw angles can be obtained by using an extended Kalman filter (EKF). These rotation angles constitute a collection as $\left\{\theta_{t_i}\right\}_{i=1}^{N}$, matching with the observation time collection $\{t_i\}_{i=1}^{N}$. Secondly, we adopt the least-square method to fit the relation between $\left\{\theta_{t_i}\right\}_{i=1}^{N}$ and $\{t_i\}_{i=1}^{N}$, and best fitted lines are acquired, then the linearity degrees of the rotation angles are measured. Finally, the thresholds of the linearity degrees are set, and intervals with higher linearity degrees than thresholds are selected as the optimum CPIs. Since the effect of the noise on the tracking data is much smaller than that on echo signal, and most of the noise can be removed by the EKF effectively, the algorithm would provide robust performance with the precise tracking information compared with the existing methods.

The rest of this paper is organized as follows. The ISAR imaging geometry and the signal model are given in section II. In section III we introduce the method of selecting the optimum CPIs; and the experimental results with both simulated and measured data are shown in section IV. In section V, some conclusions are drawn.

II. Imaging Geometry and Signal Model

The relation between Doppler frequency and the target's roll, pitch and yaw has been clearly analyzed by Chen and Miceli [34], and a brief analysis of the imaging geometry and signal model will be made in the following content.

Let us consider the ISAR imaging geometry shown in Fig.1. The target is placed in the local coordinate system (X_s, Y_s, Z_s) which is embedded on the target body with the origin O_s at the target mass center. The (X, Y, Z) is the radar reference coordinate system which acts as the absolute reference system with O being the origin. Without loss of generality, we assume that the (X_s, Y_s, Z_s) coincides with (X, Y, Z) at the starting time. R_a denotes the distance between radar and the target. The target is observed by the radar at elevation angle β and azimuth angle α at the starting time of the considered imaging interval.

Under the start-stop approximation, the received signal of ISAR can be viewed as a function of two variables, the fast time τ and the slow time t. The part of signal with respect to the fast time τ contributes to the range resolution via the use of wide bandwidth transmitted waveforms, while the part related to the slow time t contains the information of the target's relative 3-D rotational motion which determines the effective rotation vector in ISAR imaging and therefore the cross-range resolution of the reconstructed image. For convenience, the rest of this paper is mainly focused on analyzing the part of the received signal related to the slow time t. The received radar signal with respect to the slow time t can be expressed as

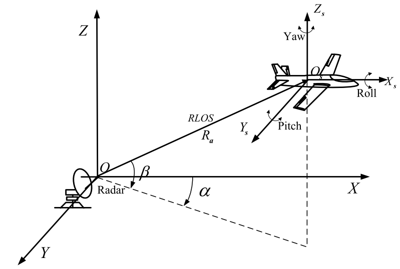

Fig. 1. ISAR imaging geometry of maneuvering target.

$$s(t) = \sum_{k=1}^{N} \sigma_k \cdot \exp\left[-j\frac{4\pi}{\lambda}R_k(t)\right] \tag{1}$$

where N denotes the number of the scatterers in a certain range bin; σ_k is the amplitude of scatterer k; λ is the wavelength; $R_k(t)$ represents the distance between the kth scatterer and the radar. In general, the instantaneous range between radar and scatter k can be expressed as

$$R_k(t) = R_a(t) + r_k(t) \tag{2}$$

where $R_a(t)$ corresponds to the translational motion part; $r_k(t)$ refers to the rotational motion part which can be expressed as equation Eq.(3) using a rotation matrix $\mathbf{Rot}(t)$.

$$r_k(t) = [\mathbf{Rot}(t) \cdot \mathbf{s}_k]^T \cdot \mathbf{i}_{los} \tag{3}$$

Where $\mathbf{s}_k$ is the position vector of point k in target body coordinate; $\mathbf{i}_{los} = [\cos\beta\cos\alpha, \cos\beta\sin\alpha; \sin\beta]^{\mathrm{T}}$ is unit vector along the radar line-of-sight (LOS); $\mathbf{Rot}(t)$ can be expressed as follows

$$\mathbf{Rot}(t_m) = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \tag{4}$$

where

$$\begin{cases} a_{11} = \cos\theta_p(t)\cos\theta_y(t) \\ a_{12} = -\cos\theta_p(t)\sin\theta_y(t) \\ a_{13} = \sin\theta_p(t) \\ a_{21} = \sin\theta_r(t)\sin\theta_p(t)\cos\theta_y(t) + \cos\theta_r(t)\sin\theta_y(t) \\ a_{22} = -\sin\theta_r(t)\sin\theta_p(t)\sin\theta_y(t) + \cos\theta_r(t)\cos\theta_y(t) \\ a_{23} = -\sin\theta_r(t)\cos\theta_p(t) \\ a_{31} = -\cos\theta_r(t)\sin\theta_p(t)\cos\theta_y(t) + \sin\theta_r(t)\sin\theta_y(t) \\ a_{32} = \cos\theta_r(t)\sin\theta_p(t)\sin\theta_y(t) + \sin\theta_r(t)\cos\theta_y(t) \\ a_{33} = \sin\theta_r(t)\cos\theta_p(t) \end{cases} \tag{5}$$

where $\theta_r(t)$, $\theta_p(t)$ and $\theta_y(t)$ represent target's roll, pitch and yaw angle, respectively. In a short imaging time interval, the variations of roll, pitch, yaw are very small (about 3°), according to which, $\mathbf{Rot}(t)$ can be approximated as [32],

$$\mathbf{Rot}(t) \approx \begin{bmatrix} 1 & -\theta_y(t) & \theta_p(t) \\ \theta_y(t) & 1 & -\theta_r(t) \\ -\theta_p(t) & \theta_r(t) & 1 \end{bmatrix} \tag{6}$$

The returned radar signal after translational motion compensation can be expressed as :

$$\begin{aligned} s(t) &\approx \sum_{k=1}^{N} \sigma_k \exp\left(-j\frac{4\pi}{\lambda} r_k(t)\right) \\ &\approx \sum_{k=1}^{N} \sigma_k \exp\left\{-j\frac{4\pi}{\lambda}[\mathbf{Rot}(t)\cdot \mathbf{s_k}]^{\mathrm{T}} \cdot \mathbf{i}_{los}\right\} \end{aligned} \tag{7}$$

On the other hand, the roll angle, pitch angle and yaw angle can be expressed as the sum of the linear and the nonlinear parts as following:

$$\begin{cases} \theta_r(t) \approx \omega_r t + \xi_r(t) \\ \theta_p(t) \approx \omega_p t + \xi_p(t) \\ \theta_y(t) \approx \omega_y t + \xi_y(t) \end{cases} \tag{8}$$

where ω_r, ω_p and ω_y are the constant angular velocities of the linear part, and $\xi_r(t)$, $\xi_p(t)$ and $\xi_y(t)$ are the nonlinear part. Substituting (8) into (6), the rotation matrix can be re-expressed as

$$\mathbf{Rot}(t) \approx \mathbf{W}t + \varepsilon(t) + \mathbf{I} \tag{9}$$

where

$$\mathbf{W} = \begin{bmatrix} 0 & -\omega_y & \omega_p \\ \omega_y & 0 & -\omega_r \\ -\omega_p & \omega_r & 0 \end{bmatrix} \tag{10}$$

$$\varepsilon(t) = \begin{bmatrix} 0 & -\xi_y(t) & \xi_p(y) \\ \xi_y(y) & 0 & -\xi_r(t) \\ -\xi_p(t) & \xi_r(t) & 0 \end{bmatrix} \tag{11}$$

where $\mathbf{I}$ is the identity matrix. Substituting (9) into (7), we have the phase of the scatterer as Eq.(12)

$$\begin{aligned} \phi_{d,s}(t) \approx -\frac{4\pi}{\lambda}[\mathbf{W}\cdot \mathbf{s_k}]^{\mathrm{T}} \cdot \mathbf{i}_{los} \cdot t - \frac{4\pi}{\lambda}[\varepsilon(t)\cdot \mathbf{s_k}]^{\mathrm{T}} \cdot \mathbf{i}_{los} \\ -\frac{4\pi}{\lambda} \cdot \mathbf{s_k}^{\mathrm{T}} \cdot \mathbf{i}_{los} \end{aligned} \tag{12}$$

where we make the approximation that the change of $\mathbf{i}_{los}$ during the imaging interval is sufficiently small and then it can be treated as a constant vector. The first term in Eq.(12) is the linear part of the phase which corresponds to the uniform rotation of the target and reflects the cross-range information after Fourier transform. The second term is the nonlinear part of the phase corresponding to the non-uniform rotation of the target, which will blur the image. The third term is a constant which has no influence on the focusing quality of ISAR image.

If the nonlinear parts of roll, pitch and yaw angles are sufficiently small, they can be treated as zeros, i.e., $\varepsilon(t) \approx 0$. Then the Eq.(12) can be re-expressed as

$$\phi_{d,s}(t) \approx -\frac{4\pi}{\lambda}[\mathbf{W}\cdot \mathbf{s_k}]^{\mathrm{T}} \cdot \mathbf{i}_{los} \cdot t - \frac{4\pi}{\lambda} \cdot \mathbf{s_k}^{\mathrm{T}} \cdot \mathbf{i}_{los} \tag{13}$$

where we can see that the phase of the scatterer is a linear function of slow time t. Then calculate the first derivative of the phase, we can get the Doppler frequency of the scatterer given by

$$f_{d,s}(t) \approx \frac{1}{2\pi}\frac{d\phi_{d,s}(t)}{dt} = -\frac{2}{\lambda}[\mathbf{W}\cdot \mathbf{s_k}]^{\mathrm{T}} \cdot \mathbf{i}_{los} \tag{14}$$

Note that the Doppler frequency remains unchanged in Eq.(14), it reflects the position information on the Doppler axis after Fourier transform.

However, if the nonlinear parts of roll, pitch and yaw angles are too large to be ignored, the Doppler frequency of the scatterer can be written as

$$f_{d,s}(t) \approx -\frac{2}{\lambda}[\mathbf{W}\cdot \mathbf{s_k}]^{\mathrm{T}} \cdot \mathbf{i}_{los} - \frac{2}{\lambda}\frac{d[\varepsilon(t)\cdot \mathbf{s_k}]^{\mathrm{T}}}{dt} \cdot \mathbf{i}_{los} \tag{15}$$

which makes the Doppler frequency change with slow time t and leads to blurred images after Fourier transform.

As analyzed above, the variety of the Doppler frequency is caused by both uniform and non-uniform change of the target's 3-D rotational motion, to put it another way, the Doppler frequency changes corresponding to the variety of target's roll, pitch and yaw motion. Therefore, instead of choosing the imaging interval by analyzing scatterers' Doppler frequency, we can select the intervals within which the roll, pitch and yaw angles have approximated linear variation as the optimal CPIs. In the following contents, we introduce a novel way to obtain target's orientation, namely, target's roll, pitch and yaw motion, and a novel selection method based on measuring the linearity degree of target's orientation.

III. Optimum CPIs Selection

A. Target Orientation Estimation

Ehrman and Lanterman [35] proposed a method for estimating the orientation of an aerial target based on velocity measurements by developing an extended Kalman filter (EKF) framework, which is a convenient way for us to estimate the target's pose accurately.

As depicted in Fig.1, yaw is defined to be the rotation about the vector coming out the top of the target. It is measured relative to the radar reference coordinate system (X, Y, Z) and modeled as a function of the target motion in the $X-Y$ plane with the scope of $0° - 360°$. Yaw is expressed by

$$\theta_y = \arctan(\frac{v_y}{v_x}) \tag{16}$$

where v_x and v_y are the velocities along X and Y, respectively. A yaw of 0° corresponds to the target flying along the positive direction of X, i.e., $v_y = 0$ in Eq.(16).

The target's pitch describes the angle between the total velocity vector and the velocity vector in the $X-Y$ plane, at which the altitude of the target changes. Under the coordinated flight model, the pitch can be approximated as

$$\theta_p = \arctan(\frac{v_z}{\sqrt{v_x^2 + v_y^2}}) \tag{17}$$

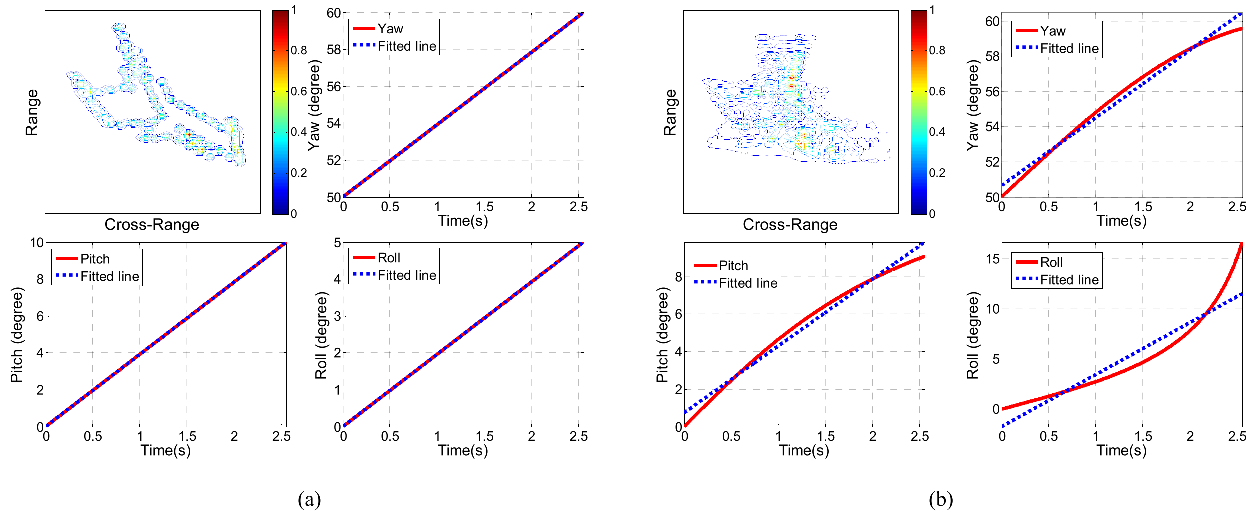

Fig. 2. The linearity degree in different time intervals. (a) In time intervals T_1, high linearity degree. (b) In time intervals T_2, low linearity degree.

Roll is the rotation about the vector extending through the target's nose, and can be approximated by

$$\theta_r = \arctan(\frac{|\upsilon|^2 \cos(\theta_p)}{Rg}) \tag{18}$$

where υ is the total velocity of the target and g is the gravity at earth's surface; R is the radius of the curvature of the target's turn, and computed as

$$R(t) = \frac{\left[\upsilon_x(t)^2 + \upsilon_y(t)^2\right]^{3/2}}{\upsilon_x(t)\, a_y(t) - a_x(t)\, \upsilon_y(t)} \tag{19}$$

where υ_x and υ_y are the velocities along X and Y, a_x and a_y are the acceleration along X and Y.

Considering two distinct types of errors (measurement error and modeling error) presenting in the estimation of target's orientation, L.M. Ehrmanand A.D. Lanterman introduced an EKF to balance both types of errors and output the target's orientation accurately. The state model and the measurement model of the EKF are described by

$$\begin{cases} \mathbf{x}_t = \mathbf{x}_{t-1} + \mathbf{q}_t \\ \mathbf{z}_t = \mathbf{h}_t(\mathbf{x}_t) + \mathbf{s}_t \end{cases} \tag{20}$$

where $\mathbf{x}_t$ is a vector consisting of the target's yaw, pitch and roll at time t; $\mathbf{q}_t$ is the process noise term, modeled as a zero-mean Gaussian random variable with covariance $\mathbf{Q}_t$; $\mathbf{z}_t$ is the measurement vector consisting of target's velocities along X, Y and Z; $\mathbf{h}_t(\mathbf{x}_t)$ is a nonlinear mapping of the state into the measurement space and $\mathbf{s}_t$ is the measurement noise term, modeled as a zero-mean Gaussian random variable with covariance S_t. More details about the filter can be found in [35].

When the tracking data, namely the distance between radar and the target $R_a(t)$, the elevation angle $\beta(t)$ and the azimuth angle $\alpha(t)$ (as shown in Fig.1) are given, the three-dimensional coordinates of the target in radar reference coordinate can be obtained by the following equation:

$$\begin{cases} x(t) = R_a(t) \cdot \cos\beta(t) \cdot \cos\alpha(t) \\ y(t) = R_a(t) \cdot \cos\beta(t) \cdot \sin\alpha(t) \\ z(t) = R_a(t) \cdot \sin\beta(t) \end{cases} \tag{21}$$

where $\left[\, x(t)\ y(t)\ z(t) \,\right]^{\mathrm{T}}$ denotes the position vector of the target in radar reference coordinate at time t. Then the velocities can be obtained by:

$$\begin{cases} \upsilon_x(t) = \dfrac{x(t) - x(t-1)}{\Delta t} \\ \upsilon_y(t) = \dfrac{y(t) - y(t-1)}{\Delta t} \\ \upsilon_z(t) = \dfrac{z(t) - z(t-1)}{\Delta t} \end{cases} \tag{22}$$

where $\left[\, \upsilon_x(t)\ \upsilon_y(t)\ \upsilon_z(t) \,\right]^{\mathrm{T}}$ is the velocity vector of the target at time t, Δt is the pulse repetition interval (PRI).

As described above, treating the velocity vector as the input of the extended Kalman filter, we can get the roll angle $\theta_r(t)$, pitch angle $\theta_p(t)$ and yaw angle $\theta_y(t)$ of the target as the output of the filter. Then, these rotation angles can be used in the following section to select the optimal CPIs.

B. Measuring Linearity Degree of Rotation Angle

To demonstrate the basic concept of linearity degree of rotation angle, we consider two time intervals, T_1 and T_2. Where, in interval T_1, target's orientation, namely roll, yaw and pitch all show a linear variation, as shown in Fig. 2 (a). Whereas in interval T_2, the three rotation angles vary non-linearly, as shown in Fig. 2 (b). The least-square linear fitting method is adopted here to find the best fitted lines of roll, pitch and yaw angles in these two time intervals, as shown in Fig. 2, the solid lines represent actual rotation angles and the dotted lines are the fitted lines. Assume that the relation between the time and fitted line is expressed as

$$\theta^{fit}(t_i) = kt_i + \theta_0 \tag{23}$$

where $\{t_i, i = 1, 2, \ldots, N\}$ indicates the vector of time, $\theta^{fit}(t_i)$ indicates the rotation angles corresponding to t_i after

TABLE I
SOME PARAMETERS OF SIMULATED RADAR SYSTEM

Center frequency	5.52 GHz	**PRF**	400 Hz	**Observation time**	2.56 s
Bandwidth	400 MHz	**Sampling frequency**	10 MHz	**Pulse width**	10 μs

linear fitting. In Eq.(23), two parameters k and θ_0 are obtained by minimizing ε where

$$\varepsilon = \sum_{i=1}^{N} \left| \theta\left(t_i\right) - \theta^{fit}\left(t_i\right) \right|^2 \tag{24}$$

where $\theta\left(t_i\right)$ is target's actual rotation angle obtained by EKF, $\theta^{fit}\left(t_i\right)$ is target's angle after linear fitting. Define the linearity degree of target's rotation angle as r, it can be measured as follows

$$r = \frac{\sum_{i=1}^{N} \left(\theta\left(t_i\right) - \bar{\theta}\right) \cdot \left(\theta^{fit}\left(t_i\right) - \bar{\theta}^{fit}\right)}{\sqrt{\sum_{i=1}^{N} \left(\theta\left(t_i\right) - \bar{\theta}\right)^2 \cdot \sum_{i=1}^{N} \left(\theta^{fit}\left(t_i\right) - \bar{\theta}^{fit}\right)^2}} \tag{25}$$

where $\bar{\theta}$ is the average value of $\theta\left(t_i\right)$; $\bar{\theta}^{fit}$ is the average value of $\theta^{fit}\left(t_i\right)$.

C. Optimum CPIs Selection

As analyzed in section II, Doppler frequency of the scatterers is composed of three components: the linear part caused by the uniform rotation of the target, the nonlinear part caused by the non-uniform rotation of the target and the constant part which has no influence on the imaging. So the time intervals within which the target mainly do uniform rotation are selected as the optimum CPIs.

After we get the target's roll, pitch and yaw angles, the linearity degree of the three rotation angles are measured over different time intervals, and the optimal CPIs can be selected. These intervals with higher linearity degree indicate that the target mainly do uniform rotation, and they are suitable for ISAR imaging. It should be pointed out that, as we have analysed in section II, the Doppler frequency is determined by roll, pitch and yaw motion jointly, an optimal CPI should be qualified with high linearity degree for all the three rotation angles. Let the observation time be T, width of the time window be T_w and the step length of the window be $\frac{T_w}{2}$. The major steps of selecting optimum CPIs are presented as follows.

- Step 1: Initialization. Let the time window be $t \in [0, T_w]$.
- Step 2: For each time window, find the best fitted lines of roll, pitch and yaw angles via the least-square linear fitting method and calculate the linearity degree of the three rotation angles respectively. Mark them as R_r, R_p and R_y.
- Step 3: Given Thr_r, Thr_p and Thr_y as the linearity degree thresholds of roll, pitch and yaw. Check whether all the linearity degree have reached the thresholds. If yes, go to Step 4. If not, slide the time window by $\frac{T_w}{2}$ and then go back to Step 2.

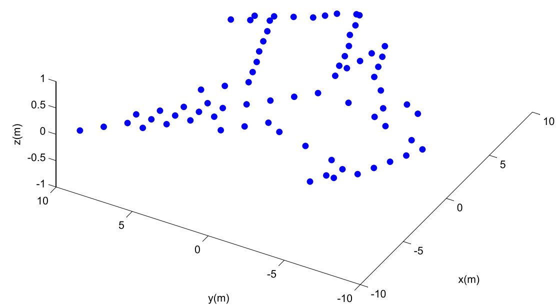

Fig. 3. Scatters distribution of simulated maneuvering target.

- Step 4: Fix time window's start time t_{start}, increase the window's termination time by PRI each time. Repeat Step 2 until $R_r = Thr_r$, $R_p = Thr_p$ or $R_y = Thr_y$ and mark the termination time as t_{end}. Update the time window to $t \in [t_{end}, t_{end} + T_w]$ and go back to Step 2.
- Step 5: Repeat Step 2-4 until the time window reaches the end of the observation time. Record all the time intervals $[t_{start}, t_{end}]$ obtained by step 4.

After the selection process, the whole observation time will be divided into several parts, including the optimum CPIs.

IV. PERFORMANCE ANALYSIS

In this section, the performance of the proposed algorithm will be evaluated by using both simulated data and real measured data. For comparison, ISAR imaging time selection method using DCF [28] is implemented in simulation experiment. What's more, different thresholds are set in the simulation experiment to analyze their influences on the performance of the proposed method, and the impact caused by the performance of the EKF is also discussed. In the simulation experiment, the scattering model of the target is distributed in Fig.3. Some radar system parameters are listed in Table.I, and the target's velocities in simulation are $v_x = 40 + 5\sin(2\pi t)$, $v_y = 50 + 5\cos(2\pi t)$ and $v_z = 2 + 4\sin(2\pi t)$, respectively.

Firstly, the proposed method in this paper is implemented to selected optimal CPIs. The target's orientation obtained by EKF is shown in Fig. 4. As we can see, the roll, pitch and yaw angles randomly change during the observation time. The method proposed in section III is applied to find the fitted lines and to measure the linearity degree at different time intervals. Width of the time window is set to 0.3s (120 pulses) and the step length is therefore 0.15s (60 pulses). The linearity degree thresholds of yaw, pitch and roll are all set to 0.99. After the selection process, the whole observation time is divided into 4 parts as shown in Fig. 4. The red dotted lines are the fitted lines after linear fitting. As we can see clearly that the roll, pitch and yaw change linearly during time interval ①

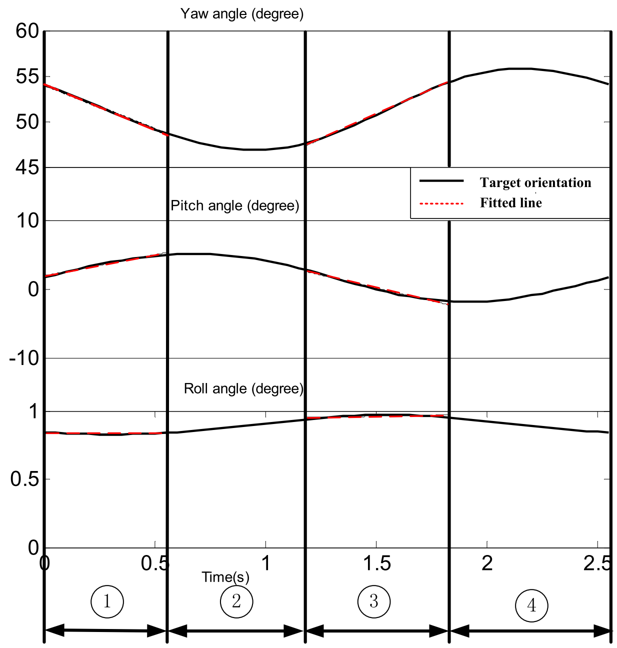

Fig. 4. Target's orientation and the division of the observation time.

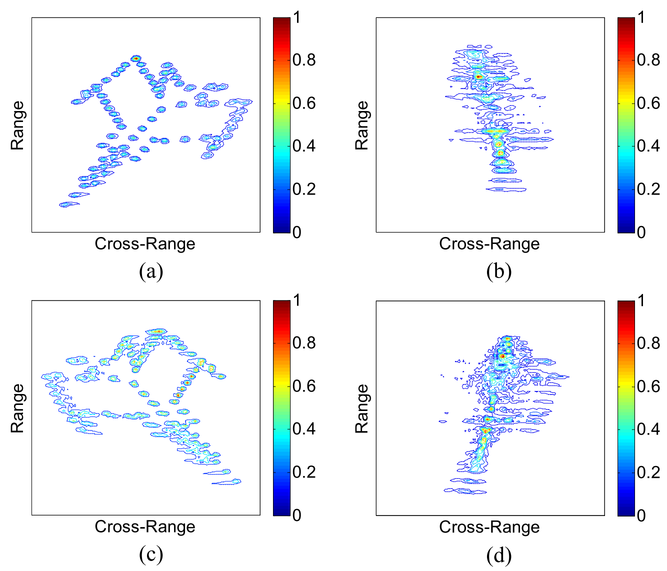

Fig. 5. The imaging results of different time intervals. (a) Time interval ①. (b) Time interval ②. (c) Time interval ③. (d) Time interval ④.

(0s-0.555s) and time interval ③ (1.18s-1.83s). Table II gives the linearity degree of the three rotation angles in different time intervals. From Table II, the linearity degree of yaw in time interval ② and interval ④ are much smaller than that of others, which would lead to nonlinear variation of phase and eventually lead to blurred images. The imaging results obtained by the range Doppler (R-D) algorithms are shown in Fig. 5. Apparently, the imaging results in time interval ① and ③ are much better than that of others.

For comparison, the ISAR imaging time selection method proposed in [28] is also adopted here to select the optimal CPIs. In [28], the DCF is obtained to estimate the variation of the Effective Rotation Vector (ERV), the optimal imaging time instants are determined by locating extreme values of

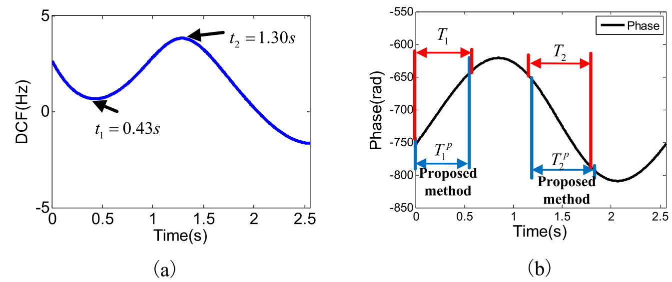

Fig. 6. The DCF curve and the selected time intervals by these two method. (a) The DCF curve. (b) The time-variant phase and the selected time intervals.

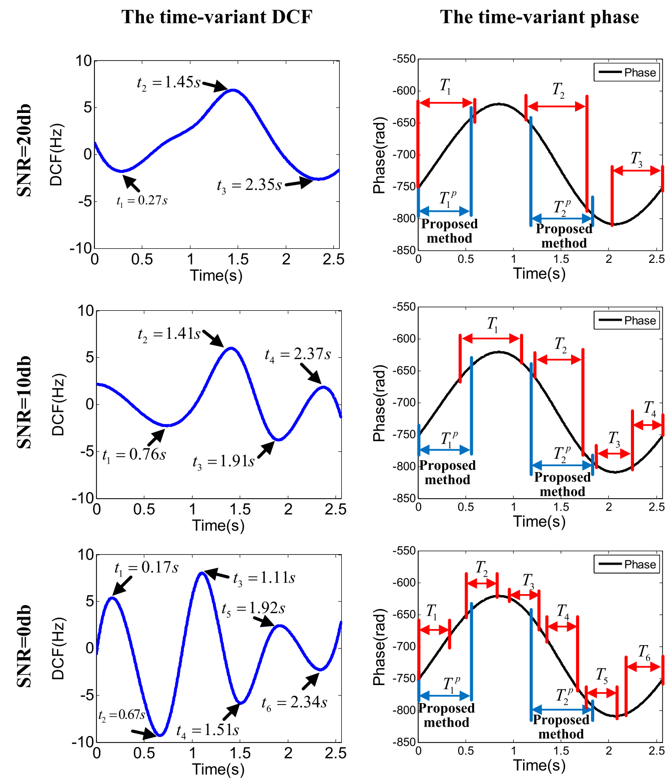

Fig. 7. The DCF curves and the selected time intervals under different SNRs.

TABLE II

THE LINEARITY DEGREE OF THE ROTATION ANGLES IN DIFFERENT TIME INTERVALS

Linearity degree / Time interval / Target's orientation	①	②	③	④
Yaw	0.9997	0.6820	0.9971	0.2583
Pitch	0.9921	0.9251	0.9920	0.9670
Roll	0.9920	0.9971	0.9938	0.9985

the estimated DCF curve where the ERV stays constant and the target rotate uniformly. The optimal durations centered with the chosen instants are determined by maximizing the contrast of the resulting images. The time varying DCF is shown in Fig.6(a), and the imaging instants selected by this method are marked with arrows. As we have discussed in section II, a well-focused ISAR image can be obtained during

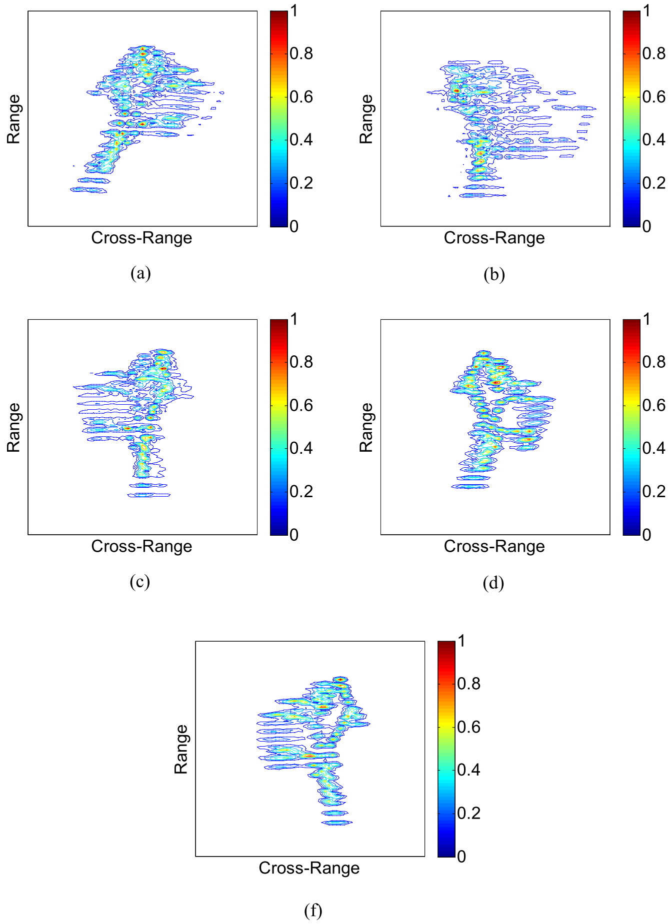

Fig. 8. The imaging results during the time intervals with low PLD selected by using DCF under different SNRs. (a) SNR=20db, $T_3 \in [2.03s, 2.56s]$, PLD=0.9662. (b) SNR=10db,$T_1 \in [0.44s, 1.08s]$, PLD=0.7223. (c) SNR=10db, $T_3 \in [1.75s, 2.25s]$, PLD=0.7311. (d) SNR=0db, $T_2 \in [0.51s, 0.83s]$, PLD=0.9788. (f) SNR=0db, $T_5 \in [1.76s, 2.08s]$, PLD=0.9630.

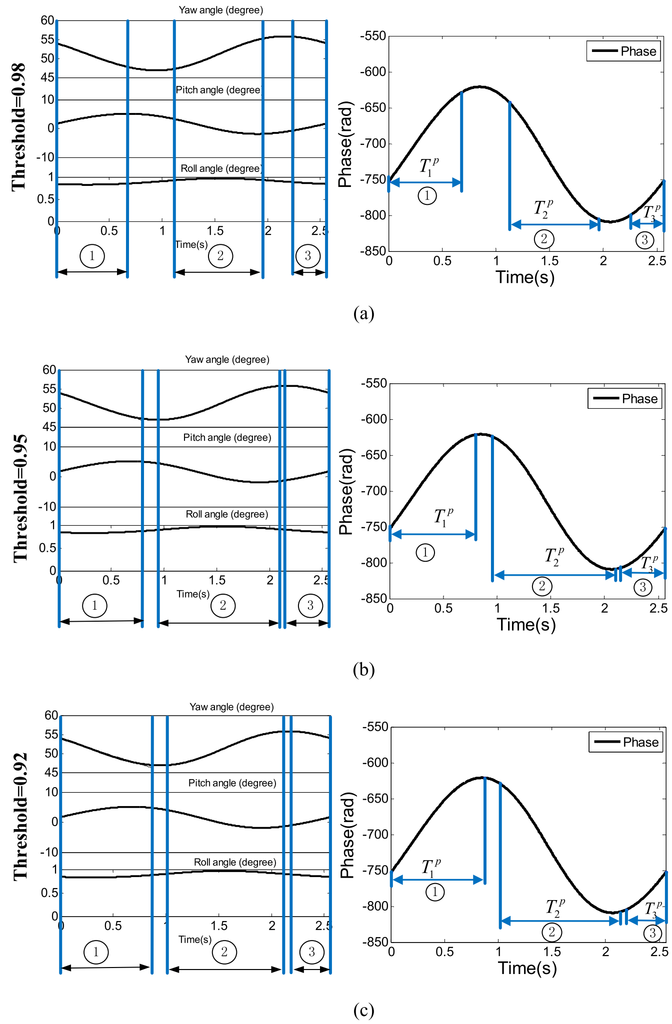

Fig. 9. The selected time intervals by proposed method using different thresholds. (a) Threshold = 0.98. (b) Threshold = 0.95. (c) Threshold = 0.92.

the time intervals within which the phase of the scatterers have approximated linear variation, so it is reasonable to evaluate the performance of the proposed method and the method using DCF by evaluating the variation characteristics of the phase in the selected time intervals. For convenient comparison, Fig.6(b) gives the time varying phase of one point on the simulated plane, and the optimal CPIs selected by these two methods are marked with blue lines and red lines on the time varying phase, respectively. The phase linearity degree (PLD) is proposed to evaluate the variation characteristics of the phase in selected time intervals. Where the PLD is defined as follows

$$\mathrm{PLD} = \frac{\sum_{i=1}^{N} \left(p\left(t_i\right) - \bar{p}\right) \cdot \left(p^{fit}\left(t_i\right) - \bar{p}^{fit}\right)}{\sqrt{\sum_{i=1}^{N} \left(p\left(t_i\right) - \bar{p}\right)^2 \cdot \sum_{i=1}^{N} \left(p^{fit}\left(t_i\right) - \bar{p}^{fit}\right)^2}} \tag{26}$$

where $p\left(t_i\right)$ is the phase value at time t_i $(i = 1, 2, \ldots, \mathrm{N})$, N is the length of the selected time interval, $\bar{p}$ is the average value of $p\left(t_i\right)$; $p^{fit}\left(t_i\right)$ is the phase after linear fitting, $\bar{p}^{fit}$ is the average value of $p^{fit}\left(t_i\right)$. The intervals with higher PLD are more suitable for ISAR imaging. As Fig.6(b) shows, T_1, T_2 are the selected time intervals by using DCF, T_1^p and T_2^p are the selected time intervals by proposed method, as we can note, with the absence of the noise, the method using DCF has almost the same performance as the proposed method, both these two methods select the time intervals within which the phase have approximated linear variation, and the PLD of the selected time intervals are $\mathrm{PLD}(T_1) = 0.9992$, $\mathrm{PLD}(T_2) = 0.9994$, $\mathrm{PLD}(T_1^p) = 0.9992$ and $\mathrm{PLD}(T_2^p) = 0.9993$. The high value of PLD proves the effectiveness of these two methods. However, as we have discussed before, the method using DCF relies on the quality of the received data, and it is sensitive to the noise. In the following, white Gaussian noise is added to the simulated data to generate different SNRs (from 20db to 0db), the DCF obtained from data with different SNRs are show in the left column of Fig.7 and the selected time intervals by these two methods are shown in the right column of Fig.7. From Fig.7 we can see that, as the SNR decreases, more and more extreme values appear, the phase in some selected time intervals are not linear variation. To show the performance of these two methods under different SNR, Table III gives the PLD values of the selected time intervals. From Table III we can see

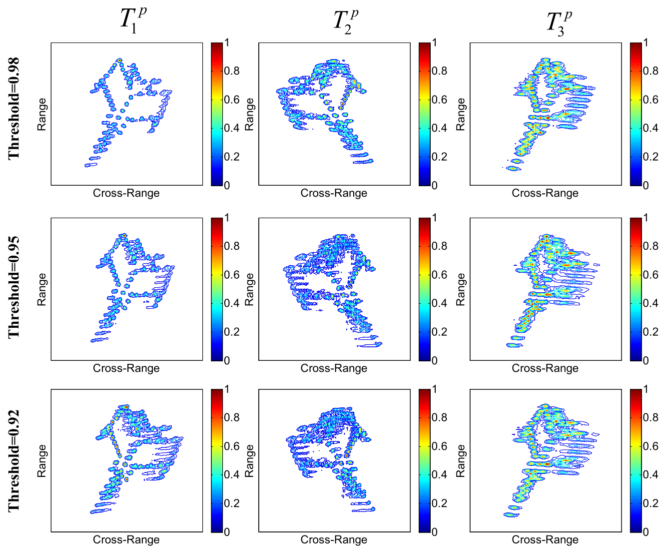

Fig. 10. Imaging results corresponding to different thresholds.

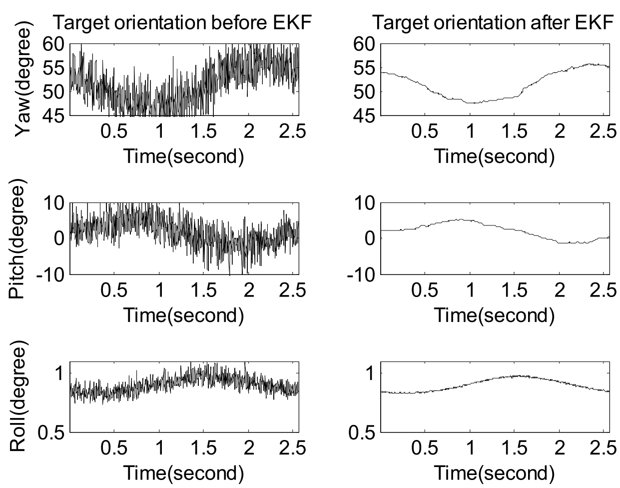

Fig. 11. Target's orientation before and after EKF using simulated data.

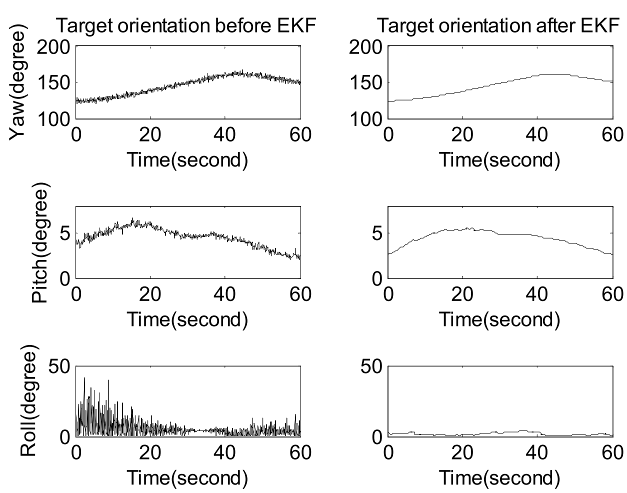

Fig. 12. Target's orientation before and after EKF using measured data.

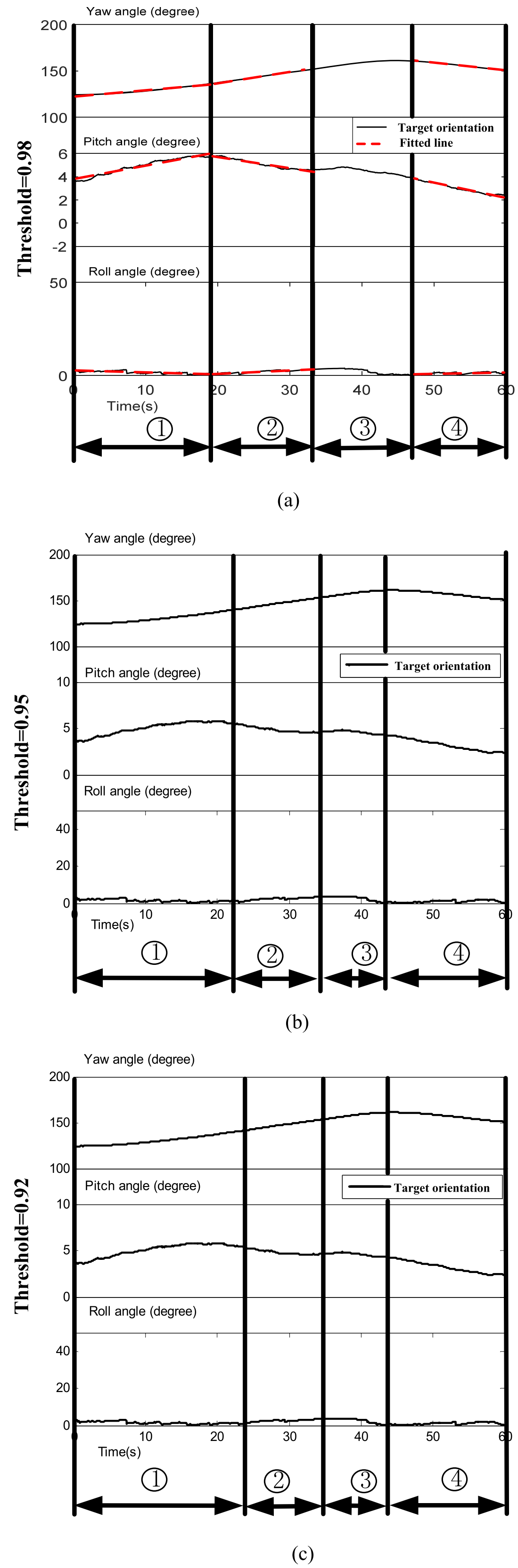

Fig. 13. Target's orientation and the division of the observation time. (a) Thresholds = 0.98. (b) Thresholds = 0.95. (c) Thresholds = 0.92.

that, as the SNR decreases, the performance of the method using DCF declines as well, many time intervals with low PLD values are selected by this method. The reason lies in that, as the enhancement of noise, the estimated DCF may be distorted by the noise, and accurate results may not be obtained by this method. On the contrary, the proposed method only

TABLE III
THE PLD VALUES OF THE SELECTED TIME INTERVALS BY THESE TWO METHODS UNDER DIFFERENT SNRS

Method	SNR	PLD					
		T_1 (T_1^p)	T_2 (T_2^p)	T_3	T_4	T_5	T_6
Proposed Method	20db 10db 0db	0.9992	0.9993	\	\	\	\
Method using DCF	20db	0.9988	0.9994	0.9662	\	\	\
	10db	0.7223	0.9989	0.7311	0.9962	\	\
	0db	0.9998	0.9788	0.9894	0.9998	0.9630	0.9919

TABLE IV
THE PLD VALUES OF THE SELECTED TIME INTERVALS BY PROPOSED METHOD USING DIFFERENT THRESHOLDS

PLD / Time intervals / Thresholds	T_1^p	T_2^p	T_3^p
0.98	0.9970	0.9962	0.9963
0.95	0.9906	0.9874	0.9883
0.92	0.9804	0.9788	0.9792

relies on the tracking data, the selected time intervals will not be influenced by the low quality of the received data, which provides robust performance in selecting the optimal CPIs. Fig.8 gives the imaging results during the time intervals with low PLD selected by using DCF. As we can see, these images are all seriously blurred, which indicates the declined performance under low SNR.

Considering the selecting steps of the proposed method present in section III, the performance of the proposed method is mainly influenced by two factors: the thresholds and the performance of the EKF. In the following contents, different thresholds will be set and the performance of the EKF will be evaluated by adding noise to the tracking data.

Firstly, three thresholds are set to the same value from 0.92 to 0.98, the selected time intervals using different thresholds are marked in the right column of Fig.9, Table IV gives the PLD values of these selected time intervals. From Table IV we can see that, the PLD values of similar time intervals decrease with thresholds decreasing. And to evaluate the performance of the proposed method using different thresholds, the ISAR images of selected time intervals are given in Fig.10. As Fig.10 shows, the quality of the images declines with the thresholds decreasing. So, in order to obtain well-focused ISAR images, the thresholds should be set above 0.98. Then, the noise is added to the target's velocities and the SNR is 0db. Target's orientation before and after filtering are shown in Fig.11. Target's yaw, pitch and roll are shown in the three rows, respectively. And the orientation before filtering is shown in the left column, the orientation after filtering is shown in the right column. As one can note from Fig.11, even the noise is added, the EKF still exhibits excellent performance as smooth results can be obtained. On the other hand, although the smooth results can be obtained, the linearity degree may be smaller than the real values when the tracking data is interfered with noise. Therefore, in order to select the optimal CPIs, the thresholds should be adjusted to smaller values compare with the empirical values given in pervious content.

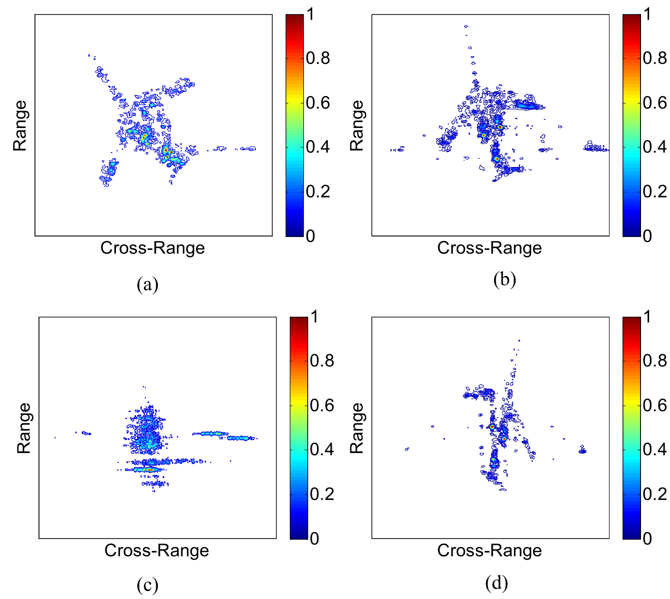

Fig. 14. The imaging results of different time intervals. (a) Time interval ①. (b) Time interval ②. (c) Time interval ③. (d) Time interval ④.

In the experiments using real measured data, the measured ISAR data is recorded by a ground-based radar for a time interval of about 60s. The parameters of the radar system are the same as that of simulated data. The same as we do to simulated data, target's velocities are calculated by using Eq.(21) and Eq.(22) at first, and then treating the velocities as the input of the EKF, target's yaw, pitch and roll can be obtained. Fig.12 gives target's orientation before and after filtering. The target's orientation before and after processing EKF are shown in the left column and right column, respectively. As one can note that, after processing EKF, the noise is almost entirely removed, and the smooth results are obtained. Then the selecting process is applied to these smooth results. Width of the time window is set to 9s (3600 pulses) and the step length is therefore 4.5s (1800 pulses). The thresholds of yaw, pitch and roll are set to 0.98, 0.98 and 0.8 respectively. Then the whole observation time is divided into 4 parts as shown in Fig. 13(a). The red dotted lines are fitted lines after linear fitting, and we can see that the rotation angles and the fitted lines coincide with each other during time

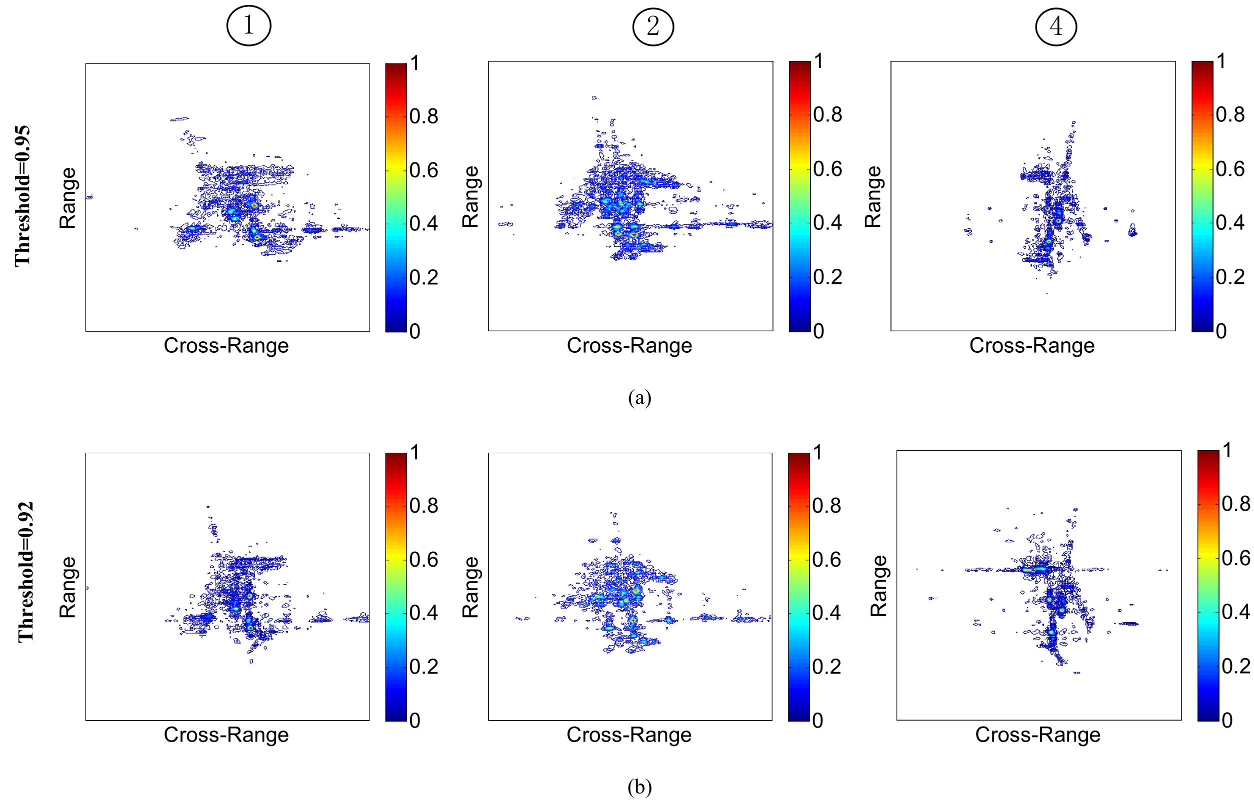

Fig. 15. Imaging results correspond to different thresholds. (a) Thresholds=0.95. (b) Thresholds=0.92.

TABLE V

THE LINEARITY DEGREE OF THE ROTATION ANGLES IN DIFFERENT TIME INTERVALS

Degree of linearity / Time interval / Target's orientation	①	②	③	④
Yaw	0.9810	0.9998	0.9496	0.9984
Pitch	0.9796	0.9793	0.8711	0.9855
Roll	0.8039	0.9120	0.3968	0.9161

interval ①, ② and ④. The linearity degrees of the rotation angles in different time intervals are listed in Table V.

The imaging results are shown in Fig. 14. We can see that, focused images can be obtained during the time interval ①, ② and ④. However, the image obtained during time interval ③ is seriously blurred. The image results show that our proposed method is also effective for measured data.

To show the influence of different thresholds on the selection results, the thresholds of the linearity degree are set to 0.95 and 0.92, respectively. The selection results corresponding to different thresholds are shown in Fig.13 (b) and Fig.13(c), and the imaging results corresponding to the selected time intervals are shown in Fig.15. From Fig.15 we can concluded that, as the thresholds drop, the quality of the images are getting worse. That is the reason we set the thresholds to 0.98. In real application, the estimated target orientation may not be very accurate, the values of the thresholds can be adjusted to meet different situations. At the end of this section, we give some empirical values of the thresholds.

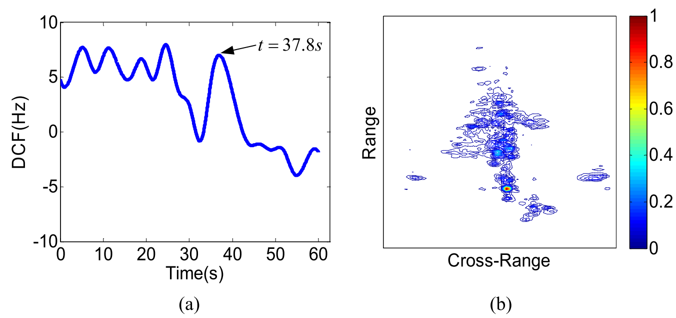

Fig. 16. The experiment results using DCF. (a) The time varying DCF and one of the selected time instants t. (b) The imaging result around time t.

Fig.16 gives the experiment results obtained by using DCF. The time varying DCF is shown in Fig. 16(a), as the algorithm suggested, time instants where the extreme values locate are selected as the optimal imaging time instants. However, due to the existence of the noise, the performance of the algorithm may decline. Fig. 16(b) gives the imaging result around $t = 37.8s$, where t is one of the extreme values shown in Fig. 16(a). As one can note, the image is very blurry, which indicates that the t is not the optimal imaging instant and the algorithm failed. In the following content, discussions on the selection process upon the experiments are given. According to the experiences in dealing with the real-measured data, we find that, in order to obtain a well focused ISAR image, the thresholds of linearity degree of the target's roll, pitch and yaw angles should be well determined in a rough scope as empirical values: $Thr_y \geq 0.95$, $Thr_p \geq 0.9$ and $Thr_p \geq 0.8$. Considering the performance of the velocity filtering, the orientation outputted by the EKF may be interfered by noise, which leads to linearity degree smaller than the real value. It is preferable to adjust the thresholds to a relatively loose scope according to the performance of velocity filtering. In addition, the thresholds of roll angle may be much nominal compared to the other rotation angles in reality. We also point out that, the width of the time window should be relatively small if the target is executing a strong maneuvering flight and relatively large if the target is executing a steady flight. But it should be long enough to obtain a desired cross-range resolution.

V. Conclusion

ISAR imaging of target is of great significance in practical applications, and however has difficulty in generating a well-focused image due to the complexity of the target 3-D motion. In this paper, a novel optimum CPIs selection method is proposed for maneuvering target. Firstly, the relation between the 3-D rotational motion (pitch, roll and yaw) and the Doppler frequency is discussed, and the target's orientation (roll, pitch and yaw) is estimated by using an extended Kalman filter (EKF), and then the linearity degree of target's orientation during different time intervals are measured, and finally the time intervals with satisfying linearity degree are chosen to be the optimum CPIs. Experiments based on both simulated data and measured data can demonstrate the effectiveness of the proposed method.

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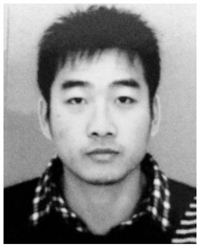

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